

SMART OR 2030



**AI-DRIVEN PATIENT AND PROCESS-SPECIFIC
PERIOPERATIVE AND SURGICAL CARE
IN THE FUTURE OR**

Convergence Health & Technology Flagship Program

Erasmus MC



 **TU Delft**



1. General information

1.a Applicants

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant]
E. Wolvius	Erasmus MC	Oral & Maxillofacial Surgery	e.wolvius@erasmusmc.nl	Tenured	Mix	Main Lead
B. Cornelissen	Erasmus MC	Oral & Maxillofacial Surgery	b.cornelissen@erasmusmc.nl	Contract until 12-'23	Researcher	Lead
E. Corten	Erasmus MC	Plastic, Reconstructive & Hand Surgery	e.corten@erasmusmc.nl	Tenured	Mix	Lead
H. Sadeghi	Erasmus MC	Cardiothoracic Surgery	h.sadeghi@erasmusmc.nl	Resident contract until 03-'24	Mix	Lead
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T. van Walsum	Erasmus MC	Radiology & Nuclear Medicine, BIGR	t.vanwalsum@erasmusmc.nl	Tenured	Researcher	Lead
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Co-applicants:

Erasmus MC			TU Delft	Erasmus University
A. Bogers	G. Tjong-A-Hung	P. Achterkamp	A. Zadpoor	C. Hilders
A. Krajnc	J. Spoor	R. Wijnen	B. Hendriks	C. Uyl-de Groot
A. Koning	J. Colaris	R. Stolker	F. Vos	M. Buljac
A. Vincent	J. Vingerling	S. Klein	F. Sterke	R. Dekker
B. Koekkoek	L. Staals	T. Piscaer	J. v Gemert	
D. Gonlag	M. Koudstaal	V. v Dal	N. Sarkalkan	
D. Gommers	M. Verhofstad	W. v Weteringen	P. Cesar	
E. v Es	M. Metselaar	W. Niessen	R. Marroquim	
E. Mahtab	M. Mureau	W. Bakhuis	X. Zhang	
E. Gundogdu	M. v Veelen	Y. Taverne		
F. Datema	P. Dekonink			
G. Roshchupkin	P. Kruizinga			

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Planned start date of seed funding: 1-10-2022

Primary Convergence theme: Human Centered Technology and AI for Health

Secondary Convergence theme: Improving Health Journeys

Keywords: Personalized Surgery, Operating Room, Artificial Intelligence, Extended Reality, 3D Printing

1.b Summary

Surgical procedures in the operating room (OR) account for up to 60 percent of a hospital's revenue and some 35-40 percent of its total expenses. However, quality of perioperative care is currently hampered by suboptimal perioperative workflows, complex patient-characteristics, and the increasing shortage of OR-personnel. Novel technologies such as computer-assisted preoperative surgical planning (e.g., using AI-powered image and data processing, virtual reality), intraoperative guidance (e.g., using augmented reality, real-time procedure monitoring, personalized 3D printed surgical guides and instruments) and feedback control systems (e.g., AI-driven algorithms to predict risk and complications, anesthesia dashboards, arterial line curve analysis) have great potential to improve the quality, safety, and efficiency of patient care in and around the operating room. Therefore, efforts are required to further develop and streamline these technologies in an efficient and safe manner targeting the best possible outcome for each individual patient.

The **SMART OR2030** Flagship aims to develop innovative technology and solutions that will optimize the perioperative workflow for complex surgical procedures. The main problem addressed by the consortium is that enormous amounts of patient-specific and process-specific data are routinely acquired in the pre-, intra-, and postoperative phases, which are not optimally being analyzed or used. Through Convergence, the SMART OR2030 consortium represents an excellent balance of research-oriented technology developers and clinical end-users with access to this procedural and patient-specific data. Moreover, SMART OR2030 has the infrastructure for

development and testing of the proposed AI-based processing solutions in real-life clinical environments. Together with the Erasmus MC AI Healthcare Hub and industrial end-product suppliers, the SMART OR2030 Flagship's demonstrators and Proof of Concept systems will be integrated in the perioperative workflows of the Erasmus MC, and will allow validation and introduction of new commercially exploitable program outputs.

1.c Public Summary

De operatiekamer (OK) levert 60 procent van de inkomsten van een ziekenhuis en omvat ongeveer 35-40 procent van de totale zorgkosten. De kwaliteit van perioperatieve zorg wordt echter geremd door suboptimale perioperatieve workflows, complexiteit van patiënten en het toenemende tekort aan OK-personeel. Nieuwe technologieën hebben een groot potentieel om de kwaliteit, veiligheid en efficiëntie van patiëntenzorg in én rond de operatiekamer te verbeteren. Bijvoorbeeld door middel van (1) computerondersteunde preoperatieve chirurgische planning (met behulp van AI-gestuurde beeld- en gegevensverwerking, virtual reality), (2) intra-operatieve begeleiding (met behulp van augmented reality, real-time procedurebewaking, gepersonaliseerde 3D-geprinte chirurgische instrumenten en implantaten) en (3) feedbackcontrole systemen (AI-gestuurde algoritmes om complicaties te voorspellen, anesthesiedashboards, arterieelijn curve analyse). Daarom zijn gecoördineerde inspanningen vereist om deze technologieën verder te ontwikkelen en te implementeren met als doel het best mogelijke resultaat voor de individuele patiënt te verkrijgen.

Het vlaggenschip **SMART OR2030** ontwikkelt nieuwe technologieën en oplossingen die complexe klinische procedures in de perioperatieve workflow van de OK's optimaliseren. Het belangrijkste probleem dat door het consortium wordt aangepakt, is de verwerking van grote hoeveelheden patiënt-specifieke en procesmatige gegevens die routinematig in de pre-, intra- en postoperatieve fasen worden verzameld. Door convergentie heeft het SMART OR2030-consortium een uitstekende balans tussen technologie- en procesontwikkelaars en klinische eindgebruikers met toegang tot deze procedurele en patiënt-specifieke gegevens. Bovendien beschikt SMART OR2030 over de infrastructuur voor het ontwikkelen en testen van AI gedreven oplossingen. Het SMART OR2030 vlaggenschip zal samen met de AI Healthcare Hub van het Erasmus MC en leveranciers van industriële eindproducten Proof-of-Concept systemen ontwikkelen. Integratie van deze systemen in de perioperatieve workflows van het Erasmus MC maken validatie en introductie van nieuwe commercieel exploitbare resultaten mogelijk.

2. Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

The vision of the SMART OR2030 flagship is to bring together a multidisciplinary cohort of scientists and clinicians in the field of data-driven surgical workflow optimization and to train early-stage researchers to the level of expertise required for designing, testing, implementing, using, managing, and translating highly innovative biomedical and computer science technology into interconnected systems for personalized surgical treatment. The flagship's network is supported by leading medical technology research groups, and academic and commercial partners with the necessary scientific and professional skill sets to translate fundamental research into technologies that can be exploited with a positive impact on socio-economic development. SMART OR2030 will ensure an excellent starting point for new scientific research lines and careers for a new generation of creative, and innovative early-stage researchers, able to face current and future challenges to design and implement process-oriented integration of technology into the next generation operating rooms (OR). SMART OR2030 will foster these long-lasting

collaborations and put Convergence Health & Technology at the forefront of research in an emerging and exciting discipline, with no parallel worldwide.

Human-centered technology & AI for health

The need for appropriate technical integration allowing for a smooth surgical workflow is not limited to the OR. Given the anticipated workforce shortages and lack of extending resources, health technologies & AI need to offer radically new ways to meet the demand of healthcare employees. This includes automation of time-consuming and error-prone perioperative processes, such as preoperative preparation, planning, and postoperative follow-up. SMART OR2030 will achieve this by the development of key technologies for **virtual surgical planning** (e.g., computer-assisted therapy preparation, 3D printing), **AI-based workflow optimization** (resource scheduling, intraoperative anesthesia and surgical monitoring), and **Extended Reality applications** to ensure workflow orchestration in and around the operating room (e.g., using AR/VR and user-specific dashboards). By developing these technologies and AI tools for predictive strategies, safety monitoring, and improved communication, we can change healthcare processes from reactive to proactive.

Improving Health Journeys

SMART OR2030 will contribute to creating a data-driven, resilient, value-based, affordable & equitable system for improved surgical procedures by using smart technologies. An automated and integrated flow reduces unnecessary workload. This frees up time to perform other necessary clinical tasks and deal with patients at a more personal level. Personalized surgery based on virtual surgical planning will result in improved accuracy of diagnosis, optimized surgical treatments. Moreover, personalized anesthesia and minimization of risks will result in improved outcome measures, decreased complication rates, and minimal side effects, making faster resumption of daily life activities possible.

2.b Synergy between disciplines

SMART OR2030 will mark an important transition point for the research field by connecting traditional disciplines of medicine with biomedical engineering, computer science, and health services research. The program draws together key researchers and departments that are essential for achieving the challenging goals of this flagship program:

- (1) Medical departments from the Erasmus MC (Anesthesiology, Cardiothoracic Surgery, Orthopedics & Sports medicine, Plastic & Reconstructive Surgery, Neurosurgery, Trauma Surgery, Oral and Maxillofacial Surgery, ENT Surgery, Pediatric Surgery)
- (2) Technical departments from the Erasmus MC (Radiology & Nuclear Medicine, BIGR)
- (3) Technical departments from the TU Delft (Biomechanical Engineering, Intelligent Systems, Imaging Physics, Applied Mathematics)
- (4) Departments from the Erasmus University (Erasmus School of Health Policy & Management, Erasmus School of Economics)

SMART OR2030 establishes a new cooperation between these departments to develop the Flagships' intertwined key technologies within common scientific, ethical, regulatory, and economic frameworks. The three key technologies will be integrated into an AI-powered platform exploiting real-life clinical data to generate an automated surgery planning and guiding pipeline. Predictive AI-enabled data analytics, developed by the technical departments (Radiology, BIGR, Intelligent systems, Applied Mathematics), will utilize pre- and intraoperative data (from the medical departments) to deliver designs (Biomechanical engineering) and quantitative insights in surgical and economic performance (EUR), which can be fed back through user-friendly XR

displays and dashboards (with industrial partners). Synergistic interactions are furthermore strongly promoted by our strategy to focus on generic technology developments for target clinical indicators that are similar across the various clinical use cases. The technical innovations will be demonstrated and evaluated in a broad range of procedures and their anesthetic management, including maxillofacial surgery, ENT-surgery, eye surgery, neurosurgery, trauma surgery, pediatric surgery, orthopedics, reconstructive surgery, and cardiothoracic surgery. Groups will study each clinical use case with different expertise, which will intensively interact, often jointly supervising their students and researchers.

2.c Long term sustainability plan

The long-term sustainability of the Flagship SMART OR2030 is highly relevant in realizing the desired impact on patients and personnel. A notable strength of the consortium is that the individual members have already initiated relevant research lines on the program's topics, which are already supported with many national and international research grants, and build upon various public-private partnerships. By integrating these research activities across disciplines and departments, SMART OR2030 will stimulate further growth and create fundraising opportunities beyond the duration and scope of currently running projects. The consortium's complementary research expertise, sharing of domain-specific knowledge, and expensive state-of-the-art infrastructures, will leverage the pace of research and fund-raising capacity, thereby catalyzing a sustainable approach for innovation within the Flagship from an early stage on.

New training and educational models

Next, to constantly seek out resources for long-term research efforts, the Flagship will sustain the work by incorporating the Flagship's developments into existing work processes and teaching models. Medical students, nurses, and residents are traditionally educated and trained in a typical clinical master-apprentice model. Technical students build up most of their experience in simulations and lab settings. However, as the complexity of technological developments in surgical care and constraints in teaching increase, it will become increasingly difficult to teach the younger generation these aspects. Embedding training for students and existing staff in a clinical research environment are examples of ways to keep up with new developments and ensure that they become part of future standard work.

SMART OR2030 Infrastructure and living lab

The partners of SMART OR2030 will collaboratively build the infrastructure and software necessary to realize the unique scientific development platform to explore and apply the technologies, best practice, methods and standards. To achieve the above, we will allocate an operating room to SMART OR2030, functioning as a living lab within the Flagship. This living lab also serves as an educational training platform, equipping today's and future students with the skills to use emerging standards and drive forward the full potential of these technological advances. Technical and medical students, residents, nurses, and surgeons of all levels can prepare, test, and perfect their developments and skills in realistic, high-tech conditions. The SMART OR will be connected to platforms like Proximie*, which combines AR and AI technology with live video streams, allowing surgeons, residents, and students to scrub in on procedures remotely.

*www.proximie.com

2.d Visibility, attracting talents and new partners

Flagship activities and events:

Scientific conferences

The SMART OR2030 network will draw on its existing links with international organizations hosting conferences in medical technology to ensure widespread awareness of the network and allow the early-stage researchers to showcase their work. Special SMART OR sessions led by the consortium members will be arranged annually at conferences including SMIT, MICCAI, CARS, SPIE MI, EMBC, AMIA, MIE, ISPOR. In addition, a final open scientific conference on surgical workflow optimization will be organized to mark the 5-year goal of the SMART OR2030 program. This conference will provide opportunities to present and discuss results with the broader scientific community and transfer key developments to stakeholders from clinical societies and hospital management.

Teaching activities

We will align with relevant existing (joint) programs, such as Technical Medicine, Biomedical Engineering, Imaging Physics, and Bioelectronics, so that students can obtain extra-curricular ECTS and allocate time to the program. Students from the EUR programs Health Science, Health Economics Policy & Law, and the European Master in Health Economics and Management (Eu-HEM) can focus their thesis topics on the societal aspects of these developments and thereby contribute to the necessary health technology assessments and long-term implementation paths. The consortium members have coordinating responsibilities in these teaching programs and a strong relationship with Avicenna Excellence Program, which attracts talents from various studies (master's and PhD students) to stimulate innovation in healthcare. Supervision will be provided by local supervisors plus an external co-supervisor from another SMART OR2030 participating organization, supporting the cross-disciplinary aspect of the research projects.

Hackathon

The SMART OR2030 hackathon will bring together students for three days of intensive collaborative work. The objective is to combine the expertise and technologies developed within SMART OR2030 in creative manners to generate innovative solutions for clinical problems, including AI and machine learning. Although the prototypes developed during the hackathon may be useful in their own right, we believe that this event's educational and social value will be of even higher interest.

SMART OR2030 summer school

We will organize SMART OR2030 summer schools that will provide the shared knowledge basis required for fluent interaction between engineering and medicine, and enhance the students' and early-stage career opportunities within and beyond the SMART OR2030 program. To create opportunities for networking beyond SMART OR2030, all summer schools will be open to students from outside the network.

Entrepreneurship Workshop

This workshop emphasizes transferable skills training: science communication (including presentation skills and outreach strategies) and entrepreneurship (including funding, team building, IP, and regulatory integration). The workshop will combine keynote seminars organized in collaboration with the SME and industry partners involved in the flagship, and will focus on lessons learned and opportunities for future development and attraction of consortium funding.

2.e Long term societal impact

Scientific impact

The complexity of operating theatres and lack of analytical methods to incorporate medical knowledge into workflow models and device automation hinder the development of intelligent assistance technologies across clinical use cases. Within SMART OR2030 we will develop machine learning-interpretable representations of clinical domain knowledge concerning perioperative protocols, OR documentation, and resource management. The modeling will be independent of concrete procedures and dynamically interlinked with multimodal data in AI-based prototype systems that provide task-specific clinical assistance in real-time. The development of context-aware medical technology will have a high scientific impact and is also exemplary for the responsible research and innovation approach of SMART OR2030, where researchers will test and validate their newly acquired methods in real-world settings.

Economic impact

OR's account for ~60% of a hospital's revenue and one OR hour costs approximately €1500-2000. However, less than 23% of all cases start on time due to inadequate alignment of activities, delays in patient transport, instrument delivery, equipment setup, and patient preparation. €420 million is lost each year in Dutch hospitals due to changeover inefficiency only. Even a 1% percentage of workflow efficiency could decrease waste by more than €1 billion worldwide. Industry participants will gain from the flagship to test novel products and solve integration challenges using emerging standards. Further, by increasing the number of highly-skilled researchers in this rapidly developing field, SMART OR2030 will strengthen the human capital base in research and innovation. Moreover, given the sustained financial pressures on healthcare providers, the positive impact of SMART OR2030 in enabling more efficient and effective use of high-cost labor and resources in surgical procedures can serve to partially reduce these financial pressures.

Societal impact

By 2025, an additional 470,000 health professionals are needed (+40%) while staff shortage is increasing rapidly. High workload, overtime, and increasing administration tasks (>53% of their time) are associated with burnout and reduced staff satisfaction. With only 27% of time spent on clinical work, patient satisfaction and safety are also under pressure. In a landmark report, the Institute of Medicine estimated that 44,000 to 98,000 deaths per year in the US were attributable to medical errors. These figures translate to 76,000 patients that suffer from unintended harm, and 2,000 deaths, for Dutch hospitals. Half of the reported adverse events are linked to disturbances and lack of awareness of unpredicted events in the perioperative process. Since optimizing efficiency and safety is a complex combinatorial problem, which is subject to a surplus of parameters and constraints, the proposed AI-based modeling and optimization techniques of this Flagship are an excellent opportunity to assist in improving and maintaining a high quality of care.

At a 5-year project level the program foresees concrete and measurable results for the following key performance indicators:

Key performance indicators	Current value	Ambition
<i>Time to access data</i>	A typical trajectory per project to get access to data for AI or analytics purposes takes between 4 weeks (fast) to 4 months.	We aim to reduce data access times in comparison to the current situation by 75%.
<i>Amount of available (high quality) data</i>	The estimate is that 90% of AI, testing and development tasks in healthcare are delayed or prohibited due to data being locked due to privacy policies and legislation.	We aim to unlock or accelerate 75% of AI, testing and development tasks targeted in the project.
<i>Performance of machine learning algorithms for perioperative optimisation of key steps</i>	Experimentally validated for a limited set of procedures. Using data input offline, optimization already reaches 70-90%	Obtain online implementation based on real-time data sources at similar or higher accuracies.
<i>Performance of machine learning algorithms for activity optimization across unobserved / untrained procedures for scalability</i>	Experimentally validated for a limited set of procedures. Using data input offline, optimization drops to ranges 4%-80%	Robust (80-90%) machine learning algorithms that can be applied on data of other types of procedures composed of similar activities.
<i>Machine learning algorithms for detecting performance gaps and risks</i>	Current limited availability data prevents automated detection of events of low incidence.	90% accurate detection of disturbances of the clinical workflow to reduce performance gaps and risks.
<i>Workload levelling: Daily OR Utilization (in %)</i>	During requirement analysis, we will compute the daily OR utilization for each room over the last 12 months and determine the planning deviations.	Reduce the variability and resulting overtime by 20%

3. Flagship proposal

3.a Ambitions

The SMART OR2030 goal: Bridging the gap of integration of technology and clinical workflow

Digitalization has improved the quality of surgery by creating a better operating environment. However, the vision of fully integrated surgical working environments, personalized process management, and the combination thereof, has not yet been realized. This obscures the increasing demand for effectiveness and efficiency of an economical and safe patient care workflow. Thus, clinicians need new tools with enhanced data-driven processing and subsequently enabled technologies such as AI-supported applications for surgical procedures. At the same time, the complexity of technical setups in an OR must remain manageable for optimal usability at the highest level. This perfectly fits with the central theme *Human-centered technology & AI for health* of Convergence Health & Technology.

SMART OR2030 advances the field of medical technologies from the manual control level to an overall automated clinical workflow integration in a wide variety of relevant clinical use cases. In doing so, the Flagship develops AI-driven patient and process-specific applications for perioperative and surgical care in the future OR.

The program focuses on the development of three key technologies in several parallel and intertwined research lines:

1. Virtual Surgical Planning

- a. Automated methods for virtual surgical planning using advanced image analysis
- b. Automated methods for designing patient-specific 3D-printed surgical guides, implants, and instruments, to enhance the surgical accuracy

2. AI-Based Workflow Optimization

- a. Next-generation feedback control systems to monitor and guide intraoperative anesthetic management to improve clinical outcomes
- b. Automated data gathering combined with AI-enabled data analytics applications to deliver real-time insight into procedural progress and risks
- c. Predictive analytics to orchestrate clinical task scheduling and patient flow, and to optimize and quantify the use of resources and human capital

3. Extended Reality Applications

- a. Augmented Reality for improved navigation and surgical guidance
- b. Virtual and Mixed Reality to enable remote presence for training and education
- c. User-specific dashboards for interdisciplinary perioperative communication

To fully unlock the potential of these key technologies, the technical developments need to be coupled with implementation research to create generalizable insights that support broad adoption and sustainability. SMART OR2030 will establish an integrative research line:

4. Evidence-Based AI Implementation Science

- a. Science frameworks for appropriate implementation strategies that promote team strengthening, leadership development, and engagement
- b. User-centered design approaches to ensure the usability of the tools
- c. Health technology assessment (HTA) using cost-effectiveness analysis

A visual summary of the SMART OR2030 program including the key technologies and implementation science is presented below:

THE ESSENCE OF THE SMART PROGRAM

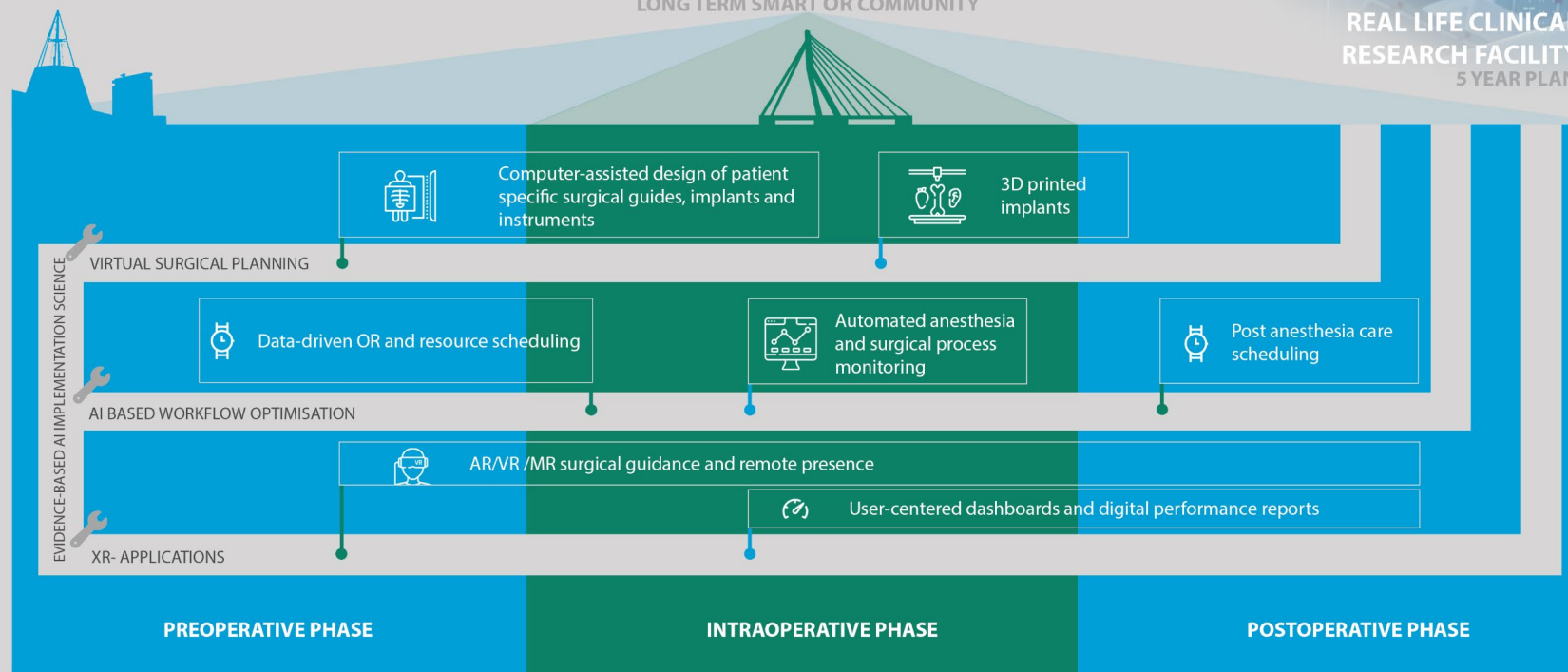
The SMART OR program is a collaboration between the Erasmus, Erasmus MC and the TU Delft. The program exists of **three key technologies**, which come into play in different projects in the three phases of surgery. By implementing the key technologies we have the aim to establish a **real life clinic research facility within 5 years**. In 2030 and beyond, we strive for a **long term smart OR community** that continuously generates ideas, tests, implements and evaluates.

SMART OR 2030

AI-DRIVEN PATIENT AND PROCESS-SPECIFIC PERIOPERATIVE
AND SURGICAL CARE IN THE FUTURE OR

LONG TERM SMART OR COMMUNITY

REAL LIFE CLINICAL
RESEARCH FACILITY
5 YEAR PLAN



SMART OR2030

Call for Flagship Programs | Convergence of Health & Technology

3.b Aims and feasibility

The key technologies delivered in the 5-year program will be developed within multiple interlinked work packages (WP) for which early-stage researchers will be appointed. The WPs have been determined based on the number of researchers, (co)financing available, and on previous/continuing collaborative projects between the participants. All early-stage researchers will interact with each other and with partners from all three institutions. BA and MSc students will participate in the WPs through internships and graduation projects. To promote collaboration, exchange of expertise, and attraction of additional funding the Flagship will organize conferences, summer schools, workshops, courses, and a hackathon. This is described in detail in 2d.

Resource and budget allocation within the program

Staff and budget of the program are allocated to the intertwined research paths, taking into account the current stage of development, the possibility to attract future additional funding, and the costs of the required infrastructure.

1. Virtual Surgical Planning

This development builds upon earlier research on automated image-based analysis for the segmentation of anatomical structures. These tools will be further developed to enable precise patient-specific surgical guides and implants (WP1: 1 PhD EMC) and instruments (WP2: 1 PD TUD).

2. AI-Based Workflow Optimization

Currently, large quantities of perioperative data are collected on clinical measurements, operative outcomes and complication rates, and available for thousands of surgical procedures. However, as it stands, only small steps in the analysis have been being taken. To boost this emerging research field, we will assign PhDs to WP3 (TUD), WP4 (EMC), and WP5 (TUD). Two will be funded through the program. One PhD (WP5) is already funded through ongoing projects.

3. Extended Reality Applications

The embedding of user-friendly XR technology in the operating room requires extensive testing and validation. Combining off-the-shelf technology (VR displays, gloves) with novel AI algorithms is needed. One PD (WP6 EMC) will work on integrating these tools across WPs.

4. Evidence-Based AI Implementation Science

One PhD will be appointed at EUR to study implementation strategies for the developed technologies. This PhD will incorporate results from test and validation studies across all other WPs within this integrative research line.

SMART OR2030 Flagship costs

Personnel costs: € 1.459.728

Material costs: € 264.500

External costs: € 785.000

Travel and publications: € 78.500

Total: € 2.587.728

Requested funding from the Convergence Health and Technology call

Total: € 1.293.864

Matching

Expected contributions from industry (see Annex support letters)

In-cash: € 280.750 ; In kind: € 715.000

Total: € 995.750

Matching from ongoing (related & funded) projects.

The consortium members already have funded projects running that fit within the scope of the program and directly contribute to the deliverables of SMART OR2030. We capitalized the yearly budget for one of these projects and included this as matching for the first two years of the program.

Total: € 312.000

Other existing external grants that fit within the scope of the program are presented at the bottom of the budget table to show our confidence in the sustainability of our project.

3.c Work plan**Work packages:**

1. Computer-assisted preoperative surgical planning
Goal: To develop 3D-printed personalized surgical guides and implants using patient-specific imaging data.
Approach: Patient-specific virtual surgical plans will provide the required input for the design of surgical devices, for a wide range of applications (e.g., pelvic bone, radius, ulna, femur, tibia, mandible, vertebrae). Surgical device designs will be supervised by surgeons, to allow monitoring and adjustments of the planning. This will make it a human-in-the-loop (HITL) application. All the adjustments will be used for continuous improvement of the algorithms to improve future designs.
Deliverables: 1) A platform where surgeons can easily create surgical devices (guides, implants) themselves. 2) Digital anatomical 3D models and virtual surgical plans can be used as input data for intra-operative navigation using augmented reality (WP6).
2. Data-driven instrument design for high precision surgery
Goal: To develop design methods based on data ranging from numerical, 2D images, 3D scans, to clinical and surgical records
Approach: We will use patient data in the development of smart instrumentation that will assist the surgeon in the precise removal of the vitreous cortex, by offering the surgeon predictive graphical overlays (WP6) clearly showing the presence of remnants, as well as by providing a dedicated instrument suited for precise removal of the remnants, e.g., by dedicated wiping. In the development, we will use 3D printed techniques (WP1) to develop novel adhesive instrument tip structures.
Deliverables: Demonstrators and prototype surgical instrumentation.
3. Data-driven prediction models to optimize OR and post anesthesia care scheduling
Goal: To develop mathematical models to optimize the day-to-day scheduling across multiple ORs and disciplines to increase OR and personnel utilization.
Approach: Several AI methods based on retrospective data will be used to develop prediction models, which will be used to optimize the preoperative daily OR schedule. By incorporating intraoperative data from WP4 and WP5, prediction of the patient's outcome will be used to develop a model (using metaheuristics and deep reinforcement learning) for the allocation of patients to postoperative destinations.

Deliverables: Trained algorithms for discipline-specific patient scheduling that can be scaled to other scheduling procedures.

4. AI algorithms for anesthesia tracking to minimize peri- and postoperative complications
Goal: To design next-generation feedback control systems to predict unfavorable events during and after the procedure and help guide good clinical practice.

Approach: Historical data pool will be utilized to build algorithms predicting outcomes, thereby guiding clinical decision-making. Data generated in the OR (e.g., arterial line waveform analysis, blood pressure measurements, medication administered and the effect on clinical parameters, ventilator settings in relation to saturation/capnography, etc.) will be used to develop AI-based systems that can inform the anesthesiologist when assessing depth of anesthesia.

Deliverables: New AI tools that blend probabilistic reasoning with deep learning. Dashboard for smooth integration of the AI-driven algorithms into the clinical workflow and to support anesthesiologists in their decision making

5. AI-enabled data to deliver real-time insight into procedural progress and risks
Goal: To develop intelligent procedure tracking to obtain real-time status updates on room level using various sensing equipment, including overview video cameras.

Approach: Automatic AI-based analysis of multimodal signals will provide insights on the progression of the procedures (for scheduling updates in WP3) and how users are interacting with the surgical tools and equipment (to identify inefficiencies and safety risks), and within the team (for task alignment and communication optimization in WP7). Tracking of positions, poses, gestures, and interactions of personnel with equipment and each other will be used as a starting point for automated data gathering.

Deliverables: Room performance metrics, improved workflow design and protocol optimization, and dashboards to increase situational awareness for the clinical team

6. Immersive technologies (VR & AR) for patient-specific preoperative planning and intraoperative guidance

Goal: To create an operational workflow for patient-specific clinical decision support using XR/immersive technologies for preoperative planning and intraoperative guidance.

Approach: Surgeons rely on their mental ability to reconstruct a realistic 3D map of vascular structures based on preoperative 2D images, memorize it, and translate it to the patient on the table. Extended Reality (XR) based on patient-specific CT-angiograms or MR-angiograms will be used to display realistic and accurate preoperative representations of complex vascular anatomical structures and to provide intraoperative real-time stereoscopic guidance.

Deliverables: User-friendly AR/VR/MR applications for intraoperative surgical guidance and remote presence for training and education.

7. Implementation research and evaluation to allow broad adoption and sustainability of novel technologies.

Goal: To couple technical advances with implementation research to create generalizable insights that support broad adoption and sustainability.

Approach: Implementation science frameworks will be developed to study how end-users interact with data and technology. Surveys and interviews will show how the surgical team functions; assess medical leadership capacity; and help to map the social networks within the department in relation to providers' knowledge, attitudes, and beliefs (KAB)

about AI. We will devise appropriate implementation strategies and study the effect of the innovations on processes and outcomes of care (e.g., patient safety, quality of care, cost-effectiveness and work satisfaction).

Deliverables: AI Implementation science frameworks and Health technology assessment (HTA) to evaluate the cost-effectiveness of the developed AI tools.

SMART OR2030 Research infrastructure and living lab

Most of the required infrastructure and equipment is already present in the host organizations, providing the researchers access to many millions of euros of specialized research facilities such as imaging systems and clinical platforms as well as pre-clinical and clinical setups for validation of the SMART OR2030 output. Specific facilities include imaging labs and experimental ORs /living lab ORs (Erasmus MC), mechanical and electrical engineering labs (TUD), software development units (all partners), and quality management systems according to current medical device regulations (all partners). The SMART OR2030 living lab will be built up within the 5-year program by interconnecting these facilities, and by the program's investments in:

- XR devices
- XR interaction devices (gloves, controllers)
- 3D OR-monitors to enable surgery in 3D
- In-house 3D printing equipment for sterilizable surgical guides, in accordance with regulatory guidelines.
- Annotation software for video and voice recognition systems
- High-performance computers and workstations with sufficient computing power to enable dynamic 3D modeling, visualization of large 3D models, and automated surgical planning.
- GPU clusters/technical staff for development and training of AI algorithms
- A dedicated OR for experimenting that can be equipped with the equipment above, and linked to platforms such as Proximie

3.d Alignment of the program with relevant national and international research agendas

The ambitions of SMART OR 2030 are in line with main national and international research agendas. Key technologies have been identified for bootstrapping the efficiency of the health care environment within roadmaps such as High-Tech Systems and Materials (HTSM), and from the European Commission (e.g. Horizon, Eureka R&D&I). The challenges include improved Healthcare Infrastructure IT for effective, massive data fusion and management, and fast extraction of the right information, and Medical Data IT which aims at the processing of medical data with use of human-centric Artificial Intelligence. Related aspects are Responsible Innovation, Ease of Use and Cooperative Systems, which have to align with hospital workflows, patient planning, patient treatment and the corresponding decision support. The current Flagship addresses these challenges and brings them together in SMART OR2030. A key result of this will be a significant increase in quality and efficiency of the healthcare system.

4. Consortium

4.a Track record of the consortium members

Flagship lead

Prof.dr. Eppo Wolvius, trained as Oral & Maxillofacial surgeon, focus on congenital facial anomalies and is leading research in the field of technology in CMF surgery. With that background, he maintains relations with various industrial partners to further embed technological development in daily clinical work. Together with other surgical disciplines and researchers from BGR he initiated Smart Surgery Lab. Recently, He has been appointed as ambassador of KTITM in the Erasmus MC. In his role as-AO CMF R&D chair of AO Foundation (the world leading organization for surgical disciplines) he oversees technological projects in the CMF field worldwide.

Leads

- Dr. B.M.W. Cornelissen, 8 years of academic research experience, yearly supervising 5-10 MSc (Technical Medicine) students.
- Dr. E.M.L. Corten, 11 years of academic experience, secured € 810.000 on external funding for projects on XR, co-guiding 4 PhD-students, supervising 3-4 Master Theses and 3-4 Bachelor Research projects (of Medical, Technical Medical & Design students) yearly.
- Dr. H. Sadeghi has >25 publications and has secured 143K of internal funding, is currently co-promotor of a PhD student, has experience in supervising several (>5) master's and bachelor students from the technical university of Delft and various medical faculties.
- Dr. J. Dauwels has won several best paper awards at international conferences and journals with his group. His academic lab has spawned four startups across a range of industries, ranging from AI for healthcare to autonomous vehicles.
- Dr. J.J. Van den Dobbelsteen >20 years of research experience, co-authored >130 journal papers, secured >11MEuro on external funding, director of studies Technical Medicine, (co)promotor of 12 Phds, current supervised research group 6 PhDs.
- Dr.ir. J-W Korstanje, 10 years of academic research experience of which 3 years as principal investigator, currently mentoring 4 PhD student, ~5 MSc/BSc students tutored every year, ~1 MEuro of external funding attracted over the last 5 years.
- Dr.ir. J.T. van Essen, 13years of academic research experience, collaborated with several hospitals, currently mentoring 4 PhD students, ~4 MSc/BSc students tutored every year.
- Dr. M.G. van Vledder has 11 years of academic research experience with 2 years as PI and has tutored many Bsc and Msc students.
- Dr. N. Ottenhof, 8 years of academic research experience, currently mentoring several MSc/BSc students.
- Prof.dr.ir. P. Breedveld, >100 MSc students, 20 PhDs, KNAW and VICI grant, involved in numerous EU and NWO projects, two of his research projects led to spin-offs.
- Dr. R. de Jong, 10 years of academic research experience, currently tutoring MSc/BSc students
- Dr.ir. T. van Walsum, 20 years expertise in image analysis for image guidance in interventions. supervised >30 BSc, >25 MSc, 12 PhDs, currently supervising 6 PhDs.
- Dr. W.K. van Deen, 10 years of academic research experience, of which 3 years as a principal investigator. Has supervised 1 PhD student and >50 MSc and BSc students. Has been awarded >€700.000 in research funding as a PI or Co-Investigator in the last 5 years.

Co-applicants

- Prof.dr.ir. R. Dekker, ranked as no. 8 in Dutch Management researcher's ranking on h-index, > 140 publications in peer reviewed journals, ERIM Impact reward, promotor of 35 PhD students, hundreds of MSc students supervised, extensive record of external funding obtained, most in combination with industry.
- Prof.dr. R-J Stolker, >30yrs of academic research experience, >100 publications in peer reviewed journals, numerous PhD students tutored, hundreds of BSc/MSc student supervised. Impressive record of accomplishment for attracting external funding.
- Prof.dr. C. Uyl-de Groot, >25 years of experience in cost-effectiveness and outcomes studies. She is supervisor of numerous PhD and MSc students has published >200 scientific articles in high impact.
- Prof.dr. Pablo Cesar leads Distributed and Interactive Systems (DIS) group at CWI. He is the PI on several EU-funded projects. In 2020, he received the prestigious Netherlands Prize for ICT Research.
- Prof.dr. C. Hilders has >10 years of clinical and research experience in medical leadership development and has supervised numerous PhD students.
- Prof.dr. Bogers, >30 years of research and clinical experience in the field of (congenital) cardio-thoracic surgery. He has co-authored more than 900 articles, has experience in supervising more than 60 PhD candidates, and has secured 18.245 million (1996-2022) of both internal and external funding.
- Dr. Martina Buljac, 15 yrs of academic research experience, has supervised 10 PhD students and >100 Bachelor and Master students. She has >30 publications and secured over €800.000 in internal and external funding.
- Dr. Mahtab, >10 years of experience in academic research and cardiothoracic surgery, has supervised 3 PhD students (amongst others a PhD on virtual reality), and is currently co-promotor of 4 PhD students. Dr. Mahtab has >40 peer-reviewed publications and has secured ±400K of both external and internal funding.
- Dr. Taverne has >10 years of academic experience and has supervised >10 PhD students in the areas of virtual reality, experimental medicine, and cardiovascular science. In addition, dr. Taverne is actively involved in multi- and transdisciplinary research and has secured ±1 million of external/internal funding.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

SMART OR2030 expertise	Virtual Surgical Planning	AI Based Workflow Optimization	Extended Reality Applications	Evidence-Based Implementation Science
Erasmus MC <i>Medical</i> Surgical, anesthesiology, and nursing departments	Virtual surgical planning for clinical applications (depts of CMF surgery, orthopedic surgery, and trauma surgery) <i>Partnership Materialise</i>	Operating room scheduling, medication administration and the effect on clinical parameters, and postoperative care scheduling. (dept of anesthesia and nurse anesthetists) <i>Partnership with Dräger</i>	Experience in the design of preoperative VR training modules. (depts of CMF surgery and cardiothoracic surgery) <i>Partnership with Medical VR</i>	Assessment of usability and evaluation of surgical outcomes
Erasmus MC <i>Technology</i> Radiology & Nuclear Medicine, BIGH	Extensive expertise in biomedical image analysis	Model-based approaches to registration and segmentation	Augmented reality for intra-operative navigation. <i>Partnership with SyncVR</i>	Assessment of usability and effectiveness
TU Delft Biomechanical Engineering, Intelligent Systems, Imaging Physics, Applied Mathematics	3D printing expertise and 3D printing facilities for the design of medical devices. <i>Partnerships with D.O.R.C and ACMIT</i>	Mathematical modelling, probabilistic reasoning and deep learning, computer vision, sensing <i>Partnerships with ACMIT and NewCompliance</i>	Graphical design of dashboards and augmented reality applications. Facilitating and improving the way people use XR technologies <i>Partnership with NewCompliance</i>	Trustworthy human-centered AI, quantitative performance metrics
Erasmus University Erasmus School of Health Policy & Management, Erasmus School of Economics	User-centered design for AI tools Patient safety and cost-effectiveness Health Tech assessment and Health Economics			Quality of care, team-strengthening and leadership development.

4.c Involvement of partners

The SMART OR2030 consortium includes several technology companies as partner organizations. For these companies, cooperation with the Flagship is intrinsically motivated by the development of key technology for AI-driven process optimization and the opening up of untapped markets for new services. Specific contributions from industry are as follows:

Materialise Medical (Leuven, Belgium) has pioneered many leading medical applications of 3D Printing and is enabling researchers, engineers, and clinicians to revolutionize innovative personalized treatment that helps improve and save lives. The company offers a flexible platform of software and services which forms the foundation of certified Medical 3D Printing of the Erasmus MC.

Involvement in Virtual surgical planning

Contribution: Cash € 100.750

Healthplus.AI (Amsterdam, Netherlands) collaborates with the Erasmus MC and uses machine learning on existing medical data to predict complications after surgery. Their first product, PERISCOPE, uses electronic health record data to predict infections after surgery. The tool is live in difference academic centers in the Netherlands, and a multicenter trial is underway to acquire CE 2a certification.

Involvement in AI Based Workflow Optimization

Contribution: Cash/in-kind € 125.000

SyncVR Medical (Netherlands) brings new Extended Reality medical inventions to prototypes that can be clinically used in the Erasmus MC. SyncVR provides healthcare institutions with a complete XR healthcare Appstore and is active in 150 healthcare organizations in 6 European countries. SyncVR Medical has an ISO27001/NEN7510 certified quality system a strong human centered design approach.

Involvement in XR Applications

Contribution: in-kind € 200.000

Medical VR (Netherlands) has been collaborating for more than two and a half year with multiple EMC departments of the Flagship. The company has co-developed and implemented Virtual Reality (VR) applications utilizing AI technology for the departments of Cardiothoracic, Oral and Maxillofacial and Plastic Surgery. The technology has also been used for training purposes of medical students.

Involvement in XR Applications

Contribution: in-kind

ACMIT GmbH (Austria) brings new medical inventions from first ideas to prototypes that can be clinically used. ACMIT also has an ISO 13485 certified quality system and has a strong human centered design approach. Together with TU Delft, ACMIT contributes to the in-depth investigation of applications in the area of "Surgical Data Science", including the use of collected data for intra-operative workflow support and self-organizing operating theatres.

Involvement in Virtual surgical planning & AI Based Workflow Optimization

Contribution: Cash €50.000, in-kind €150.000

D.O.R.C., or Dutch Ophthalmic Research Center (International B.V.) is one of world's leading companies in the field of eye surgery, based in Rotterdam/Zuidland. Within SMART OR2030, TU Delft will closely collaborate with D.O.R.C. in the design of surgical instrumentation.

Involvement in Virtual surgical planning

Contribution: Cash €30.000, in-kind €30.000

Dräger is an international leader in the fields of medical and safety technology. Dräger specializes in the care areas Intensive Care, Perioperative Care and Neonatal Care. The SMART OR2030 optimization of anastatic management through AI based feedback systems is of particular interest of the company and supported by sharing dedicated anesthesia monitoring systems.

Involvement in AI Based Workflow Optimization

Contribution: in-kind €110.000

NewCompliance is active in Operating Room departments, offering a dashboard Operating Room Cockpit, as part of a larger workflow solutions. Through AI-driven alerts and decision support developed as part of SMART OR2030, the OR Cockpit can be optimized and integrated with solutions from other participants which will increase the value (ROI) for the hospital thus improving the business case.

Involvement in AI Based Workflow Optimization & XR applications

Contribution: Cash €100.000, in-kind €100.000

4.d Governance, organization and collaboration strategy of the consortium

Organization of the consortium

The SMART OR 2030 network will be formally organized to facilitate rapid identification and resolution of issues arising and ensure effective communication across the network. The program as a whole will be led by the Program Coordinator (Erasmus MC) in collaboration with a Supervisory Board (SB) consisting of the leads from each institute, a Research and Technology Innovation Board (RTIB) with WP leaders, and the early-stage supervisors, a Career Development and Training Board (CDTB) involving academic teachers and a Clinical, Regulatory and Ethics Advisory Board (CREAB). This strategy will ensure that the network delivers an open and transparent research program in which actions are coordinated; milestones are achieved and deliverables are completed on schedule; quality standards of scientific and training procedure apply; ethical practice is maintained; activities are in concordance with the Code of Conduct of the organizations and aspects such as gender equality and diversity are meaningfully promoted.

Each WP will meet on a (bi)weekly basis and is chaired by the principal investigator (PI) of each WP to ensure the project's progress. We target the PI's of all WPs to meet every four months plenary with representatives of all parties of the consortium, where the general progress of the whole project is monitored. These meetings are chaired by the main leads of this project. Every three months all PhD, PI's/postdocs (all WPs) will have cross-over talks at separate meetings to discuss the scientific output and identify opportunities for scientific and/or societal publication of the results and transfer of knowledge to other organizations nationally and internationally.

Network communication

Effective communication will be essential to deliver the SMART OR2030 objectives. An interactive website and a confidential file-sharing site, including areas for group discussion and researcher blogs, will be established to facilitate regular online communication between researchers and partners. Web-conferencing will be extensively used for communication on both management and scientific matters. The early-stage researchers will establish monthly technical meetings to present their projects and explore opportunities for collaboration. Formal network meetings involving all researchers and associated supervisors will be held every 6 months. The network meetings will be conducted to pre-agreed agendas and include oral and poster presentations by the early-stage researchers. A separate session for the Supervisory Board (SB) will be allocated at each meeting to address any legal, ethical, contractual, financial, and administrative matters arising.

Financial management

SMART OR2030 funds will be managed by the Erasmus MC Finance Office. Detailed financial plans, including the allocation of research support costs and salary costs of early-stage researchers will be maintained by the institutes where they are appointed. The SB will ensure that the appropriate funds are dispersed to the beneficiaries according to the financial plan and implementation of deliverables. It is agreed that the coordinator will assign a Project Manager to provide network management and administrative support to the consortium.

Scientific conduct

All researchers within SMART OR2030 will be trained in and expected to observe high standards of professional behavior both in the practice and publication of their research. Any practice or conduct by a member of the SMART OR2030 team that seriously deviates from those ethical standards for proposing, conducting, and publishing research will be considered research misconduct and render them liable to disciplinary procedures of the host or the SB. Any perceived misconduct will be immediately reported to the SB and the host institution.

5. Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

Flagship:	SMART OR2030
Flagship acronym:	SMART OR2030
Naam Penvoerder:	Eppo Wolvius (Erasmus MC)
Startdatum en looptijd van het project:	1/10/2022, looptijd 5 jaar
Totaal project budget:	

1.	Loonkosten:								2022		2023		2024		2025		2026		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Inddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
	Onderzoeksorganisaties																				
	EMC	postdoc	1.0	01-10-2022	30-09-2025	Schaal 10	€ 4,802	€ 259,308	€ 18,581.25		€ 78,120		€ 90,998.00		€ 71,608		€ 259,307		€ 0	€ 259,307.36	
	EUR	PhD	1.0	01-01-2023	31-12-2026	Schaal PhD UNL	€ 3,518	€ 253,328			€ 52,001		€ 62,622.00		€ 66,800		€ 71,905.00		€ 0	€ 253,328.00	
	EMC	PhD	2.0	01-10-2022	30-09-2026	Schaal PhD EMC	€ 3,617	€ 520,891	€ 26,731.10		€ 112,384.18		€ 130,911.04		€ 139,978		€ 110,888.28		€ 0	€ 520,892.98	
	TUD	postdoc	1.0	01-10-2022	30-09-2024	Schaal 10	€ 4,802	€ 172,872	€ 18,581.25		€ 78,120		€ 76,170.57						€ 0	€ 172,872.00	
	TUD	PhD	1.0	01-10-2022	30-09-2026	Schaal PhD UNL	€ 3,518	€ 253,328	€ 13,000.25		€ 54,656		€ 63,666.50		€ 68,076		€ 53,928.75		€ 0	€ 253,328.00	
	Ondernemingen																				
	Onderneming 1						€ -	€ 0												€ 0.00	
	Onderneming 2						€ -	€ 0												€ 0.00	
	Onderneming 3						€ -	€ 0												€ 0.00	
	Onderneming 4						€ -	€ 0												€ 0.00	
	Overige (private) partners																				
	Naam 1						€ -	€ 0												€ 0.00	
	Naam 2						€ -	€ 0												€ 0.00	
	Naam 3						€ -	€ 0												€ 0.00	
	Naam 4						€ -	€ 0												€ 0.00	
	Totaal							€ 1,459,728											Totaal loonkosten	€ 1,459,728	

Kosten van materialen en hulpmiddelen:			2022		2023		2024		2025		2026		Totaal		Totaal per partner	
Partner die de kosten maakt	Materiaal / hulpmiddel	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																
EUR	Interview en enqueteskosten	€ 7,500	€ 1,500.00		€ 1,500		€ 1,500.00		€ 1,500		€ 1,500.00		€ 7,500	€ 0		€ 7,500.00
EMC	High performance computer, 3D print equipment, AR devices, 3D displays, Dashboards, s	€ 135,000	€ 40,000.00		€ 45,000		€ 20,000.00		€ 20,000		€ 10,000.00		€ 135,000			€ 135,000.00
TUD	High performance computer, sensing equipment, annotation software, design materials	€ 122,000	€ 30,000.00		€ 40,000		€ 20,000.00		€ 16,000		€ 16,000.00		€ 122,000			€ 122,000.00
Organisatie 4																€ 0.00
Ondernemingen																€ 0.00
Onderneming 1																€ 0.00
Onderneming 2																€ 0.00
Onderneming 3																€ 0.00
Onderneming 4																€ 0.00
Overige (private) partners																€ 0.00
Naam 1																€ 0.00
Naam 2																€ 0.00
Naam 3																€ 0.00
Naam 4																€ 0.00
	Totaal	€ 264,500													Totaal kosten materialen en hulpmiddelen	€ 264,500

3.	Kosten van gebruik van machines en apparatuur					2022		2023		2024		2025		2026		Totaal		Totaal per partner	
	Partner die de kosten maakt	Machine / apparatuur		Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash		in kind		in cash		in kind		in cash		in kind		
Onderzoeksorganisaties																			
EMC	Kosten gebruik Algorithm Development Server		€ 10,000	4	€ 40,000	€ 3,400.00		€ 10,000		€ 10,000.00		€ 10,000		€ 6,600.00		€ 40,000	€ 0	€ 40,000.00	
TUD	Kosten gebruik Algorithm Development Server		€ 10,000	3	€ 30,000			€ 6,600		€ 10,000.00		€ 10,000		€ 3,400.00		€ 30,000	€ 0	€ 30,000.00	
Organisatie 3																		€ 0.00	
Organisatie 4																		€ 0.00	
Ondernemingen																			
DORC	Equipment and instruments		€ 30,000	1	€ 30,000		€ 6,000		€ 6,000		€ 6,000		€ 6,000		€ 6,000	€ 0	€ 30,000	€ 30,000.00	
Healthplus AI	PERISCOPE platform		€ 25,000	5	€ 125,000		€ 25,000.00		€ 25,000.00		€ 25,000.00		€ 25,000.00		€ 25,000.00	€ 0	€ 125,000	€ 125,000.00	
DORC	Patient monitoring and IT equipment		€ 110,000	1	€ 110,000		€ 9,000		€ 27,500		€ 27,500		€ 27,500		€ 27,500	€ 18,500	€ 0	€ 110,000.00	
Sync VR	XR platform licence and engineering support		€ 200,000	1	€ 200,000		€ 10,000		€ 50,000		€ 50,000		€ 50,000		€ 40,000	€ 0	€ 200,000	€ 200,000.00	
ACMT	Development of clinical prototypes		€ 30,000	5	€ 150,000		€ 30,000		€ 30,000		€ 30,000		€ 30,000		€ 30,000		€ 150,000	€ 150,000.00	
NewCompliance	Development of clinical prototypes		€ 20,000	5	€ 100,000		€ 20,000		€ 20,000		€ 20,000		€ 20,000		€ 20,000		€ 100,000	€ 100,000.00	
Overige (private) partners																			
Naam 1																		€ 0.00	
Naam 2																		€ 0.00	
Naam 3																		€ 0.00	
Naam 4																		€ 0.00	
				Totaal	€ 785,000												Totaal kosten machines en apparatuur	€ 785,000	

[illegible]

5.

6.

Other existing external grants that are connected to the Flagship program are presented below. These grants are not included in the calculations, but show that we have confidence in the sustainability of our project.

in Euro		2022		2023		2024		2025		2026		Totaal		Totaal per partner	
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoekorganisaties															
1	EUR											€ 0	€ 0		€ 0
2	EMC											€ 0	€ 0		€ 0
3	TUD											€ 0	€ 0		€ 0
Totaal bijdrage onderzoekorganisatie		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0		€ 0
Ondernemingen															
4	Materialise	€ 20,150		€ 20,150		€ 20,150		€ 20,150		€ 20,150		€ 100,750	€ 0		€ 100,750
5	DORC	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 6,000	€ 30,000	€ 30,000		€ 60,000
6	Healthplus AI		€ 25,000		€ 25,000		€ 25,000		€ 25,000		€ 25,000	€ 25,000	€ 0	€ 125,000	€ 125,000
7	Dräger		€ 9,000		€ 27,500		€ 27,500		€ 27,500		€ 18,500	€ 0	€ 110,000		€ 110,000
8	Sync VR		€ 10,000		€ 50,000		€ 50,000		€ 50,000		€ 40,000	€ 0	€ 200,000		€ 200,000
9	MedicalVR											€ 0	€ 0		€ 0
9	ACMIT	€ 10,000	€ 30,000	€ 10,000	€ 30,000	€ 10,000	€ 30,000	€ 10,000	€ 30,000	€ 10,000	€ 30,000	€ 50,000	€ 150,000		€ 200,000
10	NewCompliance	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 100,000	€ 100,000		€ 200,000
Totaal bijdrage ondernemingen		€ 56,150	€ 100,000	€ 56,150	€ 158,500	€ 56,150	€ 158,500	€ 56,150	€ 158,500	€ 56,150	€ 139,500	€ 280,750	€ 715,000		€ 995,750
Overige (private) partners															
	Naam 1											€ 0	€ 0		€ 0
	Naam 2											€ 0	€ 0		€ 0
	Naam 3											€ 0	€ 0		€ 0
	Naam 4											€ 0	€ 0		€ 0
Totaal overige bijdrage		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0		€ 0
PPS-toeslag / subsidies / sponsors															
12	PPS-toeslag											€ 0	€ 0		€ 0
13	Philips Electronics	€ 156,000		€ 156,000								€ 312,000	€ 0		€ 312,000
14															
15															
16															
17															
Totaal subsidies		€ 156,000		€ 156,000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 312,000	€ 0		€ 312,000
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind				Total incl. PPS-toeslag
Total budget per jaar		€ 212,150	€ 100,000	€ 212,150	€ 158,500	€ 56,150	€ 158,500	€ 56,150	€ 158,500	€ 56,150	€ 139,500				€ 1,307,750
														Funding gap	1,279,978
														Requested from Convergence funds (50% of total costs)	€ 1,293,864
Totaal budget		€ 312,150	€ 370,650	€ 214,650	€ 214,650	€ 214,650	€ 214,650	€ 195,650	€ 195,650	€ 195,650	€ 195,650	€ 1,307,750			
Other existing external grants connected to the Flagship program															
	Stichting Phoenix DSW	€ 65,000		€ 65,000								€ 130,000	€ 0		€ 130,000
	Eureka R&D&I	€ 55,000		€ 55,000								€ 110,000	€ 0		€ 110,000
	Horizon2020	€ 186,000		€ 186,000								€ 372,000	€ 0		€ 372,000
	HORIZON-CL2-2021	€ 38,000		€ 38,000								€ 76,000	€ 0		€ 76,000
	ZonMW	€198,000		€198,000								€396,000	€ 0		€ 396,000
	HOKA	€123,500		€123,500								€ 247,000	€ 0		€ 247,000
Totaal subsidies		€ 665,500		€ 665,500	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 1,331,000	€ 0		€ 1,331,000

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

Addressee

University Consortium partner SMART OR2030
Institute
Mekelweg 2
2628 CD Delft



Company address

Materialise NV
Technologielaan 15
3001, Leuven
Belgium

Date: *February 28, 2022*

Subject: Support letter for Convergence Flagship ***Smart OR 2030***

Dear Prof. Wolvius ,

Materialise would like to express their warm interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

Description of company and core business

Materialise is an innovation focused company that has the expertise on bringing new medical inventions to prototypes and medical devices that can be clinically used. Materialise has an ISO 13485 certified quality system, required for this type of engineering, and has a strong human centred design approach that keeps the focus on the user value of the end product. It is Materialise's mission to innovate product development that results in a better and healthier world, through our software and hardware solutions, and an in-depth knowledge of Additive Manufacturing

Reason of interest

We believe that the proposed program targets medical needs with a strong focus on how novel technologies such as pre-operative surgical planning, intra-operative guidance and post-operative feedback loops can help to improve the quality, safety and efficiency of patient care. Materialise has been on the forefront of innovation for many years by bringing imaged based engineering and pre-operative surgical planning software together with its medical additive manufacturing services to hospitals around the world. Materialise has the ambition to continue expanding its portfolio and Smart OR 2030 fits in this strategy.

Support

If the proposal will be funded within Convergence and Materialise chooses to participate in a project, Materialise would like to participate in the development by a cash contribution of €20.150 for each year the project will be executed. Any other (in-kind) contributions can be discussed and committed based on the content of a project.

Expected new products/services

After the end of the project, Materialise will decide if the developed products can be brought to the market. The anticipated product range is a series of devices to optimize OR procedures and to improve the outcomes for the patient. It is the intent to implement the results of the project in devices that can be used within a confined OR setting for testing and validation.

Closing statements

Materialise has read the application and explicitly supports the strategy and intends to actively participate in the Flagship. By signing this letter, we confirm our intent to contribute to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

With kind regards,

Signed by an authorized signatory

Name and function

Dieter Vangeneugden
R&D Director Materialise Medical

Dieter Vangeneugden

Addressee

University Consortium partner SMART OR2030
Institute
Mekelweg 2
2628 CD Delft



Company address

D.O.R.C. Dutch Ophthalmic Research Center (International) B.V.
Kerkweg 47e,
3214 VC Zuidland
The Netherlands

Date: March, 3 2022

Subject: Support letter for Convergence Flagship *Smart OR 2030*

Dear Paul Breedveld,

D.O.R.C. would like to express their warm interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

Description of company and core business

D.O.R.C. Dutch Ophthalmic Research Center (International) B.V. is a leading manufacturing and sales company of ophthalmic equipment, instruments and liquids. Headquartered in Zuidland, The Netherlands, D.O.R.C. exports its equipment and instruments to more than eighty countries worldwide and has its own sales and marketing in the USA, Belgium, France, Germany, Scandinavia, UK, Austria, Spain, Italy, Brazil and the Middle East. DORC's vision is to deliver the best ophthalmic technology that enables surgeons to improve the quality of life for people with eye disorders.

Reason of interest

D.O.R.C. is always looking to provide the best care to the patients while giving the best surgical experience to the surgeons. D.O.R.C. strongly believes that having an OR where devices are connected together and the data coming from different sources are clustered and used in a "smart" way will help surgeons to perform the procedure in a more efficient and safe way. Improving the workflow and efficiency of eye surgery to ultimately improve the surgical patient care is exactly what brings D.O.R.C. to support the SMART OR 2030 project.

Support

If the proposal will be funded within Convergence, D.O.R.C. considers to participate in the development by a contribution in-cash in the order of magnitude of €30.000,-- over a max. period of 5 years, where the project will be executed. D.O.R.C. is further prepared to support during the course of the project with, for example, provision of equipment and instruments as well as R&D support, for an estimated total amount of €30.000,--.

The above statement has been given in good faith and to the best knowledge at the time of signature, however business priorities and economic situation may change. No legal claim can be derived from it.

Expected new products/services

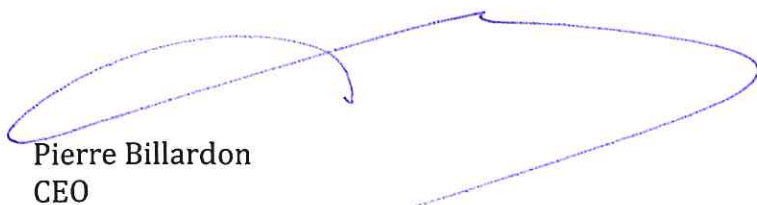
At the end of the project a series of devices and/or services to optimize OR procedures and to improve the outcomes for the patient will be developed. D.O.R.C. would help to assess the potential for bringing the developed products or services to the market and evaluate the future implementation in the OR. Discussion and agreement regarding IP will be carried out once the project is granted.

Closing statements

D.O.R.C. has read the application and explicitly supports the strategy and will actively participate in the Flagship. By signing this letter, we confirm our commitment and contribution to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

With kind regards,



Pierre Billardon
CEO

Addressee

Dr. J.J. van den Dobbelsteen
Delft University of Technology
Mekelweg 2
2628 CD Delft



Company address

Healthplus.ai-R&D B.V.
Singel 542
1017 AZ Amsterdam
The Netherlands

Date: 20-01-2022

Subject: Support letter for Convergence Flagship **Smart OR 2030**

Dear Dr van den Dobbelsteen,

Healthplus.ai would like to express their warm interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

Description of company and core business

Healthplus.ai is an expanding twelve-person startup, based in Amsterdam, that uses machine learning on existing medical data to predict complications after surgery. Our first product, PERISCOPE, uses electronic health record data to predict infections after surgery. The tool is live in LUMC and Radboud, and a multicenter trial is underway to acquire CE 2a certification under MDR early next year. We want to enable professionals to provide more proactive surgical care.

Reason of interest

We believe that the proposed program targets a serious medical need to minimize contaminations in the hospital, as contaminations do appear regular with a high financial and logistic impact on hospitals. This project fits with Healthplus.ai's focus on the surgical population and exploring/expanding the range of risk assessments we can offer to our hospital clients.

Support

If the proposal will be funded within Convergence, Healthplus.ai is considering to participate in the development by an in-kind and/or cash contribution of €25.000, for each year the project will be executed.

Expected new products/services

After the end of the project, Healthplus.ai gladly would help bring the products to the market. The anticipated product range is a series of devices to optimize OR procedures and to improve the outcomes for the patient. However, within 2 after project completion, Healthplus.ai expects to implement the initial results of the project to develop and trial SaMDs that can be used within a defined pre- and post-OR setting.

Closing statements

Healthplus.ai has read the application and explicitly supports the strategy, and will actively participate in the Flagship. By signing this letter we express our intent to contribute to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

With kind regards,

A handwritten signature in black ink, appearing to read 'B. Geerts', with a stylized, cursive script.

Bart Geerts
CEO at Healthplus.ai

Dr. N. A. Ottenhof
Erasmus Medical Center
Doctor Molewaterplein 40
3015 GD Rotterdam

March 15th 2022

LETTER OF INTENT
for the
Smart OR 2030 PROJECT

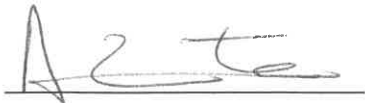
Dear Dr. N. Ottenhof,

I, Anne-Séverine Lima Pimenta, in my capacity of Clinical Engineering Manager at Draeger Medical Systems, Inc hereby confirm that Draeger Medical Systems, Inc intends to contribute to the Smart OR 2030 project, as applied for by the main applicant, Dr. Niki A. Ottenhof, anesthesiologist at the Erasmus Medical Center.

Draeger Medical Systems, Inc intends to contribute €0 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Draeger Medical Systems, Inc intends to provide an in-kind contribution of patient monitoring and IT equipment, representing a monetary value of € 110.000 and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Anne-Séverine Lima Pimenta

Position: Sr Manager of Clinical Engineering

Date: March 15th 2022

Appendix 1: in-kind contribution breakdown

Patient monitors

- 3 x Infinity Acute Care System Monitors @ 12.000 €
 - o Including patient accessories
- 2 x Device Interface Cables for Primus anaesthesia machine @ 1000 €
- 2 x Device Interface Cables for Zeus anaesthesia machine @ 1000 €
- 2 x SDC-based patient monitors @ 10.000 € (approx.) @ 10000 €

IT Support

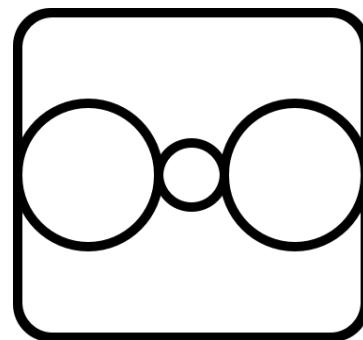
- 1 x Infinity Gateway (with software options) @ 50000 €

Addressee

University Consortium partner SMART OR2030
Institute
Mekelweg 2
2628 CD Delft

Company address

SyncVR Medical
Padualaan 8
3584CH Utrecht
The Netherlands



Date: 03-03-2022

Subject: Support letter for Convergence Flagship **Smart OR 2030**

L.S.,

SyncVR Medical would like to express their warm interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

Description of company and core business

SyncVR Medical is an innovative healthtech scale-up that has the expertise on bringing new Extended Reality medical inventions to prototypes that can be clinically used. SyncVR Medical provides healthcare institutions with a complete XR healthcare app store, currently active in 150 healthcare organisations in 6 European countries. With that, we are the largest XR platform in Europe and we have a team of 35 people. SyncVR Medical has an ISO27001/ NEN7510 certified quality system, required for this type of engineering and has a strong human centred design approach that keeps the focus on the user value of the end product. Moreover, our XR apps are CE certified as medical devices.

Reason of interest

We believe that the proposed program targets a serious medical need to minimise contaminations in the hospital, as contaminations do appear regular with a high financial and logistic impact on hospitals. SyncVR Medical is expanding their product line to medical applications and Smart OR 2030 perfectly fits this strategy.

Support

If the proposal will be funded within Convergence, SyncVR Medical would like to participate in the development by a co-finance in nature (in kind) with an amount of at least 200.000,- euro worth of business/engineering support (100 hours at a rate of €100/hour) and making our XR platform software licence available.

Expected new products/services

After the end of the project, SyncVR Medical gladly would help bring the products to the market. The anticipated product range is a series of devices to optimise OR procedures and to improve the outcomes for the patient. However, within 2 to 3 years, SyncVR Medical expects to implement the initial results of the project to develop prototype devices that can be used within a confined OR setting for testing and validation.

Closing statements

Support letter for Convergence Flagship Smart OR 2030

SyncVR Medical has read the application and explicitly supports the strategy, and will actively participate in the Flagship. By signing this letter, we confirm our commitment and contribution to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

With kind regards,

A handwritten signature in black ink, appearing to read 'Robbert Brouwer', written in a cursive style.

Robbert Brouwer
Co-founder & COO

Addressee

University Consortium partner SMART OR2030
Institute
Mekelweg 2
2628 CD Delft

Company address

MedicalVR BV
Emmalaan 7
1075AT Amsterdam
The Netherlands

Date: 01-03-2022

Subject: Support letter for Convergence Flagship **Smart OR 2030**

Dear Sir/Madame,

Hereby MedicalVR would like to express our interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

MedicalVR has been collaborating strategically with multiple Erasmus Medical Centre departments for more than two and a half years. Below explanation elaborates on how we have been co-developing and implementing Virtual Reality applications with the departments of Cardio Thoracic, Oral and Maxillofacial and Plastic Surgery.

PulmoVR our medical imaging viewer was co-invented by staff members of Cardio-Thoracic Surgery Department. The technology supports lung surgeons in complex early-stage lung cancer by utilizing Virtual Reality (VR) and Artificial Intelligence (AI) technology. The technology is structurally being implemented in a research environment at the Erasmus MC and scientific evidence has already proven clinical benefits. Research has led to publications in the Annals of Thoracic Surgery, European Heart Journal and other medical journals. MedicalVR is currently pursuing steps to obtain ISO 13485 and 27001 Certification. This technology is also being used for training purposes of Medical Students of the VECTOR-program. In addition, MedicalVR offers internships for TU Delft technical medicine students, the first has completed his 10-week internship and will stay connected to the company on a part-time basis next to his studies.

Furthermore, MedicalVR collaborates with the Oral and Maxillofacial Surgery Department on the development of surgical simulation of tools for resident surgeons and OR-assistants to better understand the steps in the procedure in VR. Within this simulator development a biomedical engineering student of the TU Delft assisted in the development of a dashboard. RTV Rijnmond made a documentary about it: [Rijnmond: Hart van Rotterdam](#)

Lastly, MedicalVR also collaborates with the Plastic Surgery Department to provide facial reconstruction patients with a better understanding of their procedure in a pre- and post-surgery setting. This multiyear public and private collaboration is partially sponsored by Top Sector Health~Holland Life Science & Health.

Considering the nature of our strategic partnerships on the beforementioned projects, MedicalVR believes in the need for a Smart OR 2030 project and supports their proposed projects. We clearly see and have faith these plans will be of great added value for the medical experts in providing better care for the patients.

In addition, we want to contribute to the SMART OR 2030 in cash and/or in-kind for multiple years once the SMART OR 2030 project goals reaches the next phase. If future prototyping and development leads to a potential medical device which could be interesting for the MedicalVR portfolio, we will support bringing the product to the market.

MedicalVR has read the application and supports the proposed strategy and projects. Comparable to the other collaborations, we will actively participate in the Smart OR Flagship project.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

Best regards,

A handwritten signature in black ink, appearing to read 'Hordijk', with a large, sweeping flourish above it.

Signed on: 01-03-2022
Chris Hordijk
Founder & CEO MedicalVR BV

ACMIT GmbH

Viktor Kaplan-Straße 2 | 2700 Wiener Neustadt | Austria
FN 273932t | LG Wiener Neustadt | www.acmit.at

Dr. J.J. van den Dobbelsteen
Delft University of Technology
Mekelweg 2
2628 CD Delft
THE NETHERLANDS

Dr. Gernot Kronreif
+43-664-5207688
Gernot.Kronreif@acmit.at

14.03.2022

Support letter for Convergence Flagship Smart OR 2030

Dear Dr. van den Dobbelsteen;

With this letter, ACMIT would like to express their warm interest in the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam.

Since 2010, the ACMIT GmbH is running an internationally oriented competence center that is mainly focused on bringing new medical inventions from first ideas to advanced, close-to-product prototypes that can be clinically used. ACMIT also has an ISO 13485 certified quality system, required for this type of engineering and has a strong human centered design approach that keeps the focus on the user value of the developed technology.

Starting with April 2022, ACMIT will further extend its scope towards in-depth investigation of applications in the area of "Surgical Data Science", including the use of collected data for intra-operative workflow support and self-organizing operating theatres. With that activity, we see a very good fit to your planned activities.

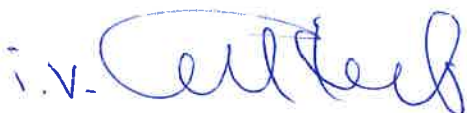
If your proposal will be funded within Convergence, ACMIT thus would like to participate in the development by a cash contribution up to €10.000,-- for each year the project will be executed. ACMIT is further prepared to help with development of clinical prototypes by a co-finance in nature (in kind) with an amount of up to €30.000,- per project year (worth of about 300 hours engineering support per year).

Both, cash and in kind project support are subject to satisfactory conditions (especially regarding access and use of jointly generated IP) to be defined in a specific contract at project start.

ACMIT has read the application and explicitly supports the strategy, and will actively participate in the Flagship. By signing this letter we confirm our commitment and contribution to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

Yours sincerely,



Dr. Gernot Kronreif
Scientific Director ACMIT

Addressee

Dr. J.J. van den Dobbelsteen
Delft University of Technology
Mekelweg 2
2628 CD Delft



Company adres

NewCompliance B.V.
Benoordenhoutseweg 46
2596 BC Den Haag

Date: *March 15, 2022*

Subject: Support letter for Convergence Flagship ***Smart OR 2030***

Dear Sir / Madam ,

NewCompliance would like to express their warm interest in and commitment to the Flagship proposal "Smart OR 2030", which you are submitting within the Flagship Program Call of Erasmus MC, Delft University and Technology and Erasmus University Rotterdam .

Description of NewCompliance and core business

NewCompliance is a medical device company that has developed a decision support platform to drastically improve safety and efficiency for Surgery and Acute Care Departments in hospitals. Our ACTIQ® platform supports caregivers with ground-breaking actionable dashboards and interoperable analytics. It is operational in more than 40 hospitals in both the EU and the US. NewCompliance has an ISO 13485 certified quality system, required for this type of engineering and has a strong human centered design approach that keeps the focus on the user value of the end product.

Reason of interest

We believe that the proposed program targets a serious medical need to minimize contaminations in the hospital, as contaminations do appear regular with a high financial and logistic impact on hospitals. NewCompliance is expanding their product line to medical applications and Smart OR 2030 perfectly fits this strategy.

Support

If the proposal will be funded within Convergence, NewCompliance would like to participate in the development by a cash contribution of €20.000, for each year the project will be executed. NewCompliance is further prepared to help with development of clinical prototypes by a co-finance in nature (in kind) with an amount of at least 100.000,- euro worth of business/engineering support (100 hours at a rate of €100/hour).

Expected new products/services

After the end of the project, NewCompliance gladly would help bringing the products to the market. The anticipated product range is a series of software applications to optimize OR procedures and to improve the outcomes for the patient. However, within 2 to 3 years, NewCompliance expects to implement the initial results of the project to develop prototype devices that can be used within a confined OR setting for testing and validation.

Closing statements

NewCompliance has read the application and explicitly supports the strategy, and will actively participate in the Flagship. By signing this letter we confirm our commitment and contribution to the program.

We are confident that this notification serves the initiative and hope for a successful collaboration in the near future.

With kind regards,

A handwritten signature in black ink, appearing to read 'Bo Wiesman', written in a cursive style.

Signed by an authorized signatory

Bo Wiesman,
CEO NewCompliance

III Letter of support from department head of the main lead

On headed paper of the department or institution.

TO WHOM IT MAY CONCERN

Doorkiesnummer 010 - 7034138

Faxnummer 010 -4633098

Intern postadres NG-309

E-mail a.s.m.janssens@erasmusmc.nl

Ons kenmerk WOL/AJA/0322005

Datum 12th March 2022

Letter of Support for Flagship SMART OR2030

Dear Madam/Sir,

I am writing in the support of the Flagship SMART OR2030 – proposal for a convergence-grant to develop innovative technology and solutions that will optimize complex clinical procedures in the perioperative workflow of the ORs. This consortium will specifically focus on patient-specific and process-specific data that are routinely being acquired in the pre-, intra-, and postoperative phases, which are not optimally being analyzed or used.

The SMART OR2030 Flagship draws together an excellent balance of technology developers and clinical end-users that will strengthen the scientific Convergence Health & Technology community within the TU Delft, Erasmus MC and Erasmus University Rotterdam. Beyond the involvement of key researchers and departments, SMART OR2030 realizes the shared infrastructure for development and testing of AI-based processing solutions in real-life clinical environments. This allows validation of the Flagship's demonstrators and introduction of new commercially exploitable program outputs in the Erasmus MC.

The synergy between our three institutions is essential for achieving the challenging goals as presented in this Flagship program. I am convinced that SMART OR2030 contributes to the Convergence Health & Technology mission and meets all requested criteria of the program convincingly. Through this letter, I acknowledge that the researchers and staff dedicated to this Flagship program will fulfill the specific roles and responsibilities in this partnership.

Looking forward to leading this Flagship SMART OR2030 wholeheartedly, and awaiting the outcome of the review process,

your sincerely,



Prof. dr. E. B. Wolvius,
Head of the Department of Oral &
Maxillofacial Surgery, Special Dental Care
and Orthodontics

Postadres

Postbus 2040
3000 CA Rotterdam

Bezoekadres

Dr. Molewaterplein 40
3015 GD Rotterdam

Contact & route

www.erasmusmc.nl

Afdelingshoofd:

Prof. dr. E.B. Wolvius

Stafleden Kaakchirurgie:

Drs. A. Gül
Dr. B.P. Jonker
Dr. M.J. Koudstaal
Drs. F.S. Kroon
Drs. H. Mast
Dr. A.V.J. Rozeboom
Drs. E.M. Strabbing
Drs. P.H. van Twisk

Onderzoek:

Dr. E.J. Farrell

Stafleden Bijz. Tandheelkunde

Drs. J.F.A. van Elswijk
Dr. S.J. van der Geer
Drs. W. van Nood
Dr. A. Tahmaseb
Drs. E.C. te Veldhuis
Drs. H.M.E. van Verseveld
Drs. C.G. Vis

Stafleden Orthodontie:

Dr. P.L. Carvajal Monroy
Drs. M. van de Lagemaat
Drs. S. Tjoa

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